

Checkpoint F4: The FTC for line integrals

Consider the vector field $\vec{G}(x, y) = \langle 3y^2 - \cos(x), 6xy + e^y \rangle$.

1. Is $\vec{G}$ a gradient field? That is, can you find a potential function $g(x, y)$ such that $\vec{G} = \nabla g$?
2. Compute $\int_C \vec{G} \cdot d\vec{r}$ where C is the top half of the unit circle oriented from $(-1, 0)$ to $(1, 0)$.